

Decision Theory and Social Choice

PE3101P — Lecture 1

Tomi Francis

National University of Singapore

13th August, 2026

Today

- 1 Logistics
- 2 Course Overview
- 3 Two senses of "ought"
- 4 Acts, states, outcomes
- 5 Chances, credences, utilities
- 6 Bayesianism

Hello

- ▶ I'm your instructor for **PE3101P: Decision Theory and Social Choice**.

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- ▶ Please call me **Tomi**.

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- ▶ Please call me **Tomi**.
- ▶ No need for "Dr. Francis" or "Prof. Francis" or anything like that.

The register

- ▶ **Please sign the register** as it comes round.

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- ▶ Add your name, signature and student number.

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- ▶ If you're not on the sheet, you won't be recorded as present (obviously).

Finding me

► **Email:** tomi.francis@nus.edu.sg

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- ▶ (My office may move later this semester. I'll tell you if it does.)

What this course is about

Decision-making in two kinds of case:

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- (1) We don't know how our actions will turn out, and must weigh the risks.

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- ▶ (1) is **decision theory**.
 - ▶ (2) is **social choice theory**.

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Roughly half the semester on each. Decision theory first.

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- ▶ Trouble getting at anything? Email me.

The syllabus

- ▶ A printed copy should be in front of you — if not, grab one from the front.

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- ▶ **Logistical question? Reread the syllabus first.** It may well answer it.

The textbook

Schwarz, *Belief, Desire, and Rational Choice* (2022)

Open access. A PDF is on my website.

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Readings

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- ▶ Listed in the syllabus — see my website.

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- ▶ Tutorials are more discussion-focused — I'll want more from you there.

Use of AI

Assessed work

Do not use AI in any way. This is cheating.

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Everything else

Use it or don't — but use your judgement. Help your learning, don't inhibit it.

Use of AI: good ideas

- ▶ Generating extra exercises based on the optional assignments.

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- ▶ Triaging extra readings you've found, to see if they're worth your time.

Use of AI: bad ideas

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In short

Do use AI to supplement your learning. **Do not** use it to replace your learning.

We'll start with tables like this

	It rains	It doesn't rain
Take the umbrella	Dry, but carrying it	Dry, but carrying it
Leave the umbrella	Soaked	Dry, hands free

- ▶ We'll learn to pick out the **states**, the **acts** and the **outcomes**.

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- ▶ We'll learn to pick out the **states**, the **acts** and the **outcomes**.
- ▶ Then **dominance principles** — and why we must be careful with them.

Decision theory: what's coming

- **Probabilities** for states of nature: what they mean, what rules they obey, and why it's irrational to break those rules.

Decision theory: what's coming

- ▶ **Probabilities** for states of nature: what they mean, what rules they obey, and why it's irrational to break those rules.
- ▶ **Newcomb's problem**: what to do when you can **expect** things to go badly if you perform a certain act, even though you **know** performing that act will make things better.

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- ▶ (If that sounds impossible — stick around.)

Decision theory: the centrepiece

Expected Utility Theory

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- ▶ i.e. as though you'd put numbers on how good each outcome is, and were making the average as high as possible.
- ▶ Then: **money pump arguments** — if you don't choose that way, you can be made worse off for no reason, **and you can see it coming**.

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- ▶ **Arrow**: the impossibility of a satisfactory voting system.
- ▶ **Is voting irrational?** You almost certainly won't change the result.

So — let's start on decision theory.

The Miners Problem

THE MINERS PROBLEM

Ten miners are trapped in a shaft, threatened by rising water. You don't know whether they are in shaft *A* or shaft *B*.

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Ten miners are trapped in a shaft, threatened by rising water. You don't know whether they are in shaft *A* or shaft *B*. You have enough sandbags to block **one** shaft, not both. Block the right shaft and all survive; block the wrong one and all die. Do nothing, and both shafts half-fill: **one miner dies**.

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	Miners in A	Miners in B
Block A	all 10 live	all 10 die
Block B	all 10 die	all 10 live
Block neither	1 dies	1 dies

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What should you do?

Three questions

- ▶ Should you block A , block B , or neither?

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- ▶ What if the miners *are* in shaft A ? What then?

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- ▶ Should you block A , block B , or neither?
- ▶ What if the miners *are* in shaft A ? What then?
- ▶ What if the miners are in shaft A **and you still don't know it?**

Two senses of “ought”

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What you ought to do is whatever will *in fact* be best. You may have no idea what that is. Never mind.

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Evidence-relative

What you ought to do is whatever is best *given your evidence*. Here, what you know matters a lot.

Which one is this course about?

Decision theory is about the evidence-relative sense.

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Decision theory is about the evidence-relative sense.

How can we decide well, on the basis of the evidence we actually have?

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The **columns** are the **states** — or *states of nature*.

- ▶ Ways the world might be that you're **uncertain** about.
- ▶ That you **can't change**.
- ▶ That **determine** what each act leads to.

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The **rows** are the **acts**.

- ▶ The choices open to you.
- ▶ Together with the state, they fix what happens.

Reading the table

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The **cells** are the **outcomes** — or *final outcomes*, or *possible worlds*.

- ▶ A complete description of a possible result of your decision.

Suppose we had a fourth option

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- ▶ vs **Block A** : better if they're in B , same if in A .

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- ▶ vs **Block *A***: better if they're in *B*, same if in *A*.
- ▶ vs **Block *B***: better if they're in *A*, same if in *B*.
- ▶ vs **Block neither**: better **whichever** state obtains.

Two kinds of dominance

Weak statewise dominance

Definitely at least as good, and **possibly** better.

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Better **no matter which state obtains**.

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So **Block both** weakly statewise-dominates blocking *A* alone, and blocking *B* alone.

Strong statewise dominance

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So **Block both** strongly statewise-dominates **Block neither**.

So should we always choose dominant acts?

It seems so.

So should we always choose dominant acts?

It seems so.

Why accept a risk — or a certainty — of a worse outcome?

A counterexample?

A student wonders whether to study for an exam.

	Will Pass	Won't Pass
Study	Pass & No Fun	Fail & No Fun
Don't Study	Pass & Fun	Fail & Fun

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Does “Don’t study” statewise dominate “Study”?

Yes — and that's the problem

- ▶ The lesson is **not** that statewise dominance is wrong.

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- ▶ It's that we set up the **states** badly.

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Remember

States of nature are things you **can't change**.

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States of nature are things you **can't change**.

But studying *does* make a difference to whether you pass.

The fix

- ▶ We should choose statewise-dominant acts — **but only once the states are set up properly.**

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- ▶ (We'll see what that means later.)

Probabilities matter too

What we should do depends not just on the states and outcomes, but on **how likely** each state is.

Credences

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- ▶ The probabilities we care about are **credences**, or *degrees of belief*.
- ▶ Roughly: how likely **you think** something is, given your evidence.
- ▶ When I say "probability", I mean **credence**, unless I say otherwise.

Credences aren't objective chances

JACK AND JILL

Jack has flipped a coin and is looking at it. It landed heads.

Credences aren't objective chances

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- ▶ Jill's **credence** in tails should be **50%**.

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JACK AND JILL

Jack has flipped a coin and is looking at it. It landed heads. Jill didn't see it, and is trying to guess.

- ▶ The **chance** of tails is **zero**. It landed heads.
- ▶ Jill's **credence** in tails should be **50%**.

We care about credences because we often don't know the chances — and we want to decide well on the evidence we actually have.

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\$100 if a fair coin lands heads, \$0 if tails. The **expected** amount is **\$50**.

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Note: the expected amount is \$50 even though **I know for certain I won't get \$50**.

Maximising expected utility

	Heads ($p = 0.5$)	Tails ($p = 0.5$)
Take the bet	\$100	\$0
Decline	\$50	\$50

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- ▶ Each row is a set of possible outcomes with probabilities attached.
- ▶ So each **act** has an **expected utility**.

It depends how you assign the utilities

Risk-neutral: every dollar worth the same

$$u(\$100) = 100, \quad u(\$50) = 50$$

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Bet: **50.** Decline: **50.** **Indifferent.**

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$$u(\$100) = 80, \quad u(\$50) = 50$$

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Bet: **40**. Decline: **50**. **Don't bet.**

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Both are ways of maximising expected utility.

One thing I haven't done

I haven't given you **any argument** that you *should* maximise expected utility.

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That comes in several weeks' time.

Which credences should we have?

We won't answer that fully. But there's a partial answer:
Bayesianism.

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Probabilism

Credences must obey the **probability axioms**.

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Probabilism

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The Principle of Conditionalisation

They must change in response to evidence by **conditionalisation**.

The probability axioms

- ▶ Write X for the set of all possible outcomes — the event that *something happens*.

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- ▶ Write X for the set of all possible outcomes — the event that *something happens*.
- ▶ An **event** is a set of possible outcomes. (Including none of them, and all of them.)

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Question: Can you show that $\text{Cr}(\sigma) + \text{Cr}(\sigma^c) = 1$?

Left for next week: Conditionalisation

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Conditional probability formula

$$\text{Cr}(A \mid B) = \frac{\text{Cr}(A \cap B)}{\text{Cr}(B)} \quad \text{provided } \text{Cr}(B) > 0.$$

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See you next week.